bb-buck — Design Document

ai-ee v1 pipeline

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Contents

1 Board and Run Metadata

Field	Value
board	bb-buck
workspace	`boards/bb-buck`
phase	P2
gate status	no gates recorded yet
state created	2026-08-15T15:58:13
state updated	2026-08-15T18:13:15

2 Overview

Brief - bb-buck (as given by the owner, verbatim)

ultra bare bones design: a switch-mode step-down converter. Input 24 V DC nominal (18-30 V) on a screw terminal; output 5 V at up to 2 A on a screw terminal. Bench supply in, resistive load out. Workspace boards/bb-buck.

Orchestrator notes (not owner words)

- Mode token present: `ultra bare bones design:` -> the ultra-bare-bones scope contract in `reference/build-modes.md` applies.
- Block under study: the synchronous/asynchronous step-down (buck) converter.
- Bench-only article: bench DC supply on the input, resistive load on the output.

Stackup

Chosen stackup: **‘JLC2313_1.6‘** - JLCPCB standard 2-layer, 1.6 mm, **1 oz outer copper**,

3 Requirements

Requirements: bb-buck

1. Function

A bare step-down (buck) DC-DC converter, built as a study article. It takes 18-30 V DC (24 V nominal) from a bench power supply on a 2-pin screw terminal and produces 5 V at up to 2 A (10 W) on a second 2-pin screw terminal feeding a resistive load. Non-isolated, common ground in to out. Nothing else is on the board: no MCU, no second rail, no digital interface, no indicators.

BUILD MODE: **ultra-bare-bones** (declared by the brief’s opening token, recorded as a P0 decision; contract in `reference/build-modes.md`). The block under study is the buck converter itself. Scope is that block plus exactly the support parts its datasheet requires at the stated operating point, one input and one output interface, and what the fab needs to build the board and the bench needs to hold it. Protection, filtering, indicators, spare rails and enclosure features are OUT by mode, not by engineering judgment. The mode’s defaults are applied below (quantity 5, JLC PCBA single-sided, bench 0-50 C indoor, no enclosure, fewest honest layers, smallest honest outline, stop at P9) instead of being asked.

Synchronous vs asynchronous rectification is deliberately NOT fixed here. The brief leaves it open and it is a P1 architecture call, not a requirement. Both are in scope as long as the block meets the stated operating point.

2. Interfaces

Exactly two, both stated by the brief.

- Power input (J1): 2-pin screw terminal (stated). DC only, 18-30 V. ASSUMED: through-hole 2-pole terminal block, 5.08 mm pitch, rated >= 300 V and >= 10 A so it is never the limiting element against the ˜0.7 A worst-case input current. Pitch was not stated; 5.08 mm is the common, well-stocked JLC/LCSC choice and accepts bench-supply lead wire comfortably. Polarity silkscreen-marked (+ / -). NOTE: the board has no reverse-polarity protection by mode, so the silk marking is the only defense against a swapped supply - see section 8.
- Power output (J2): 2-pin screw terminal (stated). 5 V, up to 2 A. ASSUMED: same terminal family and part as J1 for BOM consolidation, placed on a different board edge and silk-labelled distinctly (VIN vs 5V OUT) so the two cannot be confused at the bench.
- No other external interface exists or is wanted: no USB, no header, no comms bus, no external enable, no power-good, no fault output, no LED. All excluded by mode.
- Measurement access ANSWERED (A4): exactly one switch-node probe pad with an adjacent ground pad. No other test points - VIN, VOUT and GND are reachable at the terminals.

3. Power

- Source: bench DC power supply, 18-30 V, 24 V nominal (STATED explicitly by the brief - not a guess, not a battery, not mains-derived on this board, not automotive). Section 8 depends on this being stated.
- Load: resistive, up to 2 A at 5 V (STATED). ASSUMED: the load range is 0 to 2 A, i.e. the board must also be stable and safe with NO load connected (the realistic bench case of powering up before the resistor is attached). This is the conservative reading of ”up to 2 A” and it constrains the converter’s light-load behavior, so it is recorded rather than left silent.

- ASSUMED: 2 A is the absolute maximum output, continuous. A resistive load draws no surge beyond its steady value, so no peak allowance above 2 A is carried. Inrush into the board's own output capacitance at startup is the converter's soft-start problem, not a load requirement.

- Output rail: one rail, fixed 5 V nominal. ASSUMED: fixed output, not adjustable, no trim, no sequencing. Accuracy and ripple ANSWERED (A3): +/-3 % DC (4.85-5.15 V) across the full 18-30 V input and 0-2 A load range, ripple <= 50 mV peak-to-peak.

- Input current (GUESS, for sizing only): 10 W out at an assumed 88-93 % efficiency -> 10.8-11.4 W in. At 18 V low line ~0.60-0.63 A; at 24 V ~0.45-0.47 A; at 30 V high line ~0.36-0.38 A. Size the input terminal, input copper and input capacitor RMS rating for >= 1 A continuous with a switched peak of ~2.5 A. Output terminal and copper for >= 2.5 A continuous.

- Board dissipation (GUESS): ~0.8-1.4 W typical, ~1.8 W at a pessimistic 85 % efficiency. Worst case sits at the 30 V high-line corner where switching loss dominates. At the mode's 50 C maximum ambient with natural convection this is a real thermal driver on a small outline - copper area under the converter is a design input, not an afterthought.

- Duty cycle (derived, for P1): D ~ Vout/Vin = 0.28 at 18 V, 0.21 at 24 V, 0.17 at 30 V. The 30 V corner sets the minimum on-time constraint, which bounds how high the switching frequency can go for a given part - a real P1/P2 selection constraint, flagged here, not decided here.

- No battery, no charging, no energy storage function, no backup rail. None stated and none implied by a bench supply source.

- Excluded by mode (SCOPE decisions, not engineering conclusions - do not record any of these as generally unnecessary): input TVS/clamp, reverse polarity protection, input fuse, pi/EMI input filter, OVP/OCP/UVLO beyond what lives inside the converter IC, output LC filter beyond what the datasheet requires.

4. Environment

Mode defaults - applied, not asked.

- Ambient: 0 to 50 C, indoor bench.

- Cooling: natural convection only. No forced airflow, no heatsink, no chassis thermal path.

- No enclosure. No ingress (IP) rating. No vibration or shock requirement. No humidity requirement. No conformal coating.

- No formal EMC/emissions campaign - this is a bench study article, not a product. Tight power-loop layout is still required because it is what makes the block correct, not because a test demands it.

5. Size & mounting

- Outline: **no HARD cap.** Nothing here binds permanently at P5 board_init. The mode governs instead: the smallest outline that keeps the layout HONEST - never so tight that the power loop, the copper thermal area or the terminal spacing stop being representative of a real 24 V -> 5 V / 2 A buck, and never padded for features this board does not have.

- Non-binding expectation for planning only (NOT a cap): roughly 30-40 mm x 20-28 mm, dominated by the two screw terminals plus the converter, inductor and capacitors. If the honest layout needs more, it takes more.

- Layers: ASSUMED 2, per the mode's "fewest layers the block honestly needs". If the P2/P4 thermal work shows 2 layers cannot carry ~1.2-1.8 W at 50 C ambient inside the honest outline, 4 layers is allowed under the same rule. Not fixed here either way.
- Mounting: ASSUMED 4 x M3 clearance holes (3.2 mm) inset from the corners, so the board can sit on standoffs at the bench. Mounting the bare board is explicitly in mode scope. If the honest outline turns out too small for four, two on opposite corners is acceptable.
- Height: ASSUMED no maximum. The board is open on a bench with no enclosure, and the screw terminals themselves stand ~10-12 mm tall with wire fitted, so a height cap would be arbitrary.
- Connector placement: ASSUMED both terminals on board edges with their wire openings facing outward, on different edges, so bench wiring does not cross the board. Which edges is the layout engineer's call.

6. Quantity & budget

Mode defaults - applied, not asked.

- Build quantity: 5.
- Cost: minimal, with no hard per-unit cap. Prefer JLC Basic/Preferred parts wherever a Basic part actually meets the electrical requirement; Extended is allowed for the converter IC and the power inductor, where Basic stock is thin in this class.

7. Assembly

Mode defaults - applied, not asked.

- JLCPCB PCBA, single-sided: all SMT parts on the top side only. The bottom side stays clear of SMT so it can serve as unbroken thermal/return copper.
- The two screw terminals are through-hole. ASSUMED: fitted by JLCPCB through-hole assembly if offered at acceptable cost for the chosen part, hand-soldered after SMT otherwise. Either path leaves the schematic and the footprint unchanged.
- Pipeline stops at P9 (fab package + DFM). Ordering is a separate owner decision and is not requested here.

8. Compliance/safety flags

No safety question is open on this board, and that conclusion is derived, not assumed. The mode grants no silence here - it simply happens that the brief states the two facts that would otherwise have to be asked.

- **Mains voltage: DOES NOT APPLY.** The board sees DC only. Generating that DC is out of scope; the brief names a bench supply as the source.
- **Batteries: DOES NOT APPLY.** The source is STATED as a bench power supply. This is the one question that has no default and must never be guessed, and the brief answers it explicitly, so it is not asked. No battery on the board, no charging function, no pack chemistry, no deep-discharge or UVLO behavior required. If the owner ever runs this board from a battery instead, that is a new requirement and this section must be re-opened.
- **Motors: DOES NOT APPLY.** The load is STATED as resistive. No inductive, stalling or regenerative load, therefore no reverse energy into the output and no startup surge beyond the resistor's own draw.

- **RF transmit: DOES NOT APPLY.** No radio function.
- **High current (>3 A): DOES NOT APPLY.** Maximum continuous current anywhere on the board is the 2 A output. Input current peaks at ˜0.63 A continuous at the 18 V low-line corner. The largest instantaneous current is the inductor peak, GUESSED at ˜2.3-2.6 A depending on the ripple ratio P1 chooses - still under 3 A. The threshold is not approached closely enough to be a judgment call.
- **>30 V: DOES NOT APPLY at the stated number, but the range's top edge is genuinely under-defined.** Reasoning, explicitly:
  - The stated maximum is 30 V, and the threshold is *greater than* 30 V. 30 is not >30, so the flag does not trip on the stated figure. This is a boundary case, so it is reasoned about rather than waved past.
  - No real hazard exists at 30 V DC. It is well below the 60 V DC limit for a non-hazardous DC source under IEC 62368-1 (ES1), so there is no shock hazard to a person handling the board. Fault energy is bounded by the bench supply's own current limit, and at <= 1 A input there is no arc or energy hazard at the terminals. No safety question follows from the voltage itself.
  - Downstream consequence worth noting: at exactly 30 V, the VIN-to-GND net pair is 30 V apart, which the pipeline's `check_creepage` treats as a clean no-op (its derived check engages only above 30 V). If the answer to Open question 1 lifts the ceiling above 30 V, that check engages for real and IPC-2221 spacing starts binding the layout. The boundary is therefore not cosmetic.
  - What WAS genuinely ambiguous - whether 30 V is the *operating* or the *survival* ceiling, and what the input does on a live connection - is now ANSWERED (A1, A2): 30 V is a hard maximum OPERATING input, and there is no live hot-plug. The converter carries >= 36 V absolute-max as its own headroom; no clamp is added. The flag therefore stays DOES-NOT-APPLY on settled requirements, not merely on the boundary reading.
- **Reverse polarity: noted, not flagged.** The mode excludes reverse polarity protection, so connecting the bench supply backwards will destroy the converter. This is an accepted consequence of the declared scope on a bench article with silk-marked polarity, not a safety hazard to a person, and not a finding for a reviewer to raise. It is recorded so the consequence is visible rather than implicit.

9. Open questions - ALL FOUR ANSWERED 2026-08-15 (owner)

STATUS: **closed.** The owner took the stated DEFAULT on all four. Each answer is recorded as a P0 `state.py decision` and is now a REQUIREMENT, not a question. P1/P2 read the ANSWERED block below; the original question text is kept underneath it for provenance only.

Answers (binding)

- **A1 - Input ceiling.** 30 V is a HARD maximum operating input. Choose a converter whose absolute-maximum input rating is >= 36 V class (˜20 % headroom above 30 V). The headroom lives in the PART, not in added components - no clamp, no TVS. Consequence retained: VIN-to-GND stays at or below 30 V, so `check_creepage` remains a no-op and IPC-2221 spacing does not bind the layout.
- **A2 - No live hot-plug.** The input is never connected while the supply is live: wires land first, then the supply ramps from zero. The ˜2x lead-inductance ringing case (˜60 V from a 30 V setting) is therefore OUT of the requirement set. The datasheet-required bulk input capacitance stays fitted and damps the incidental case; no damping component is added. A 36 V-class part is sufficient under this answer.

- **A3 - Output spec.** +/-3 % DC, i.e. 4.85-5.15 V, held across the FULL 18-30 V input range and the FULL 0-2 A load range, with <= 50 mV peak-to-peak output ripple. These are the numbers the feedback divider tolerance, the output capacitance and the P8 simulation bounds must meet.
- **A4 - Probe access.** Exactly ONE switch-node probe pad with an adjacent ground pad. Nothing else: VIN, VOUT and GND are already reachable at the two screw terminals. Mode-boundary ruling by the owner - the switch node counts as the block's own measurement need on a study article. Keep the pad minimal and inside the switch-node copper that already exists, so it adds as little area as possible to the noisiest node.

Original question text (provenance) 1. **Is 30 V truly the highest the supply will ever be set to, or should the board survive a bit more?** A bench supply set to "30 V" may actually sit at 30.5 V, and a knob can be nudged. This board has no input clamp by design, so the converter chip's own maximum rating is the only thing standing between an overshoot and a dead board. DEFAULT: 30 V is a hard maximum operating input, and the converter is chosen with an absolute-maximum input rating at least ˜20 % above it (36 V class or better) purely as selection headroom. No extra parts are added. 2. **Will you ever connect the input wires while the bench supply is already switched on and at voltage?** Connecting live through a metre of lead wire makes the input voltage ring up to roughly twice the supply setting for a few microseconds - about 60 V from a 30 V supply - because the lead inductance rings against the board's ceramic input capacitors. That transient is well beyond what a 36-40 V part tolerates, and the mode excludes the clamp that would normally absorb it, so the answer changes which parts are even eligible. DEFAULT: no live hot-plug. You connect the wires first, then bring the supply up from zero. The bulk input capacitance the datasheet already requires stays fitted and damps the incidental case; nothing extra is added. If you answer "yes, I will hot-plug it", say so - it will force either a much higher-voltage part or a damping component the mode would otherwise exclude, and that is an owner call. 3. **How tight does the 5 V output need to be?** Two numbers: how far from exactly 5.00 V is acceptable, and how much high-frequency ripple can ride on top of it. DEFAULT: +/-3 % DC (4.85-5.15 V) across the full 18-30 V input range and the full 0-2 A load range, and <= 50 mV peak-to-peak output ripple. Ordinary, honest numbers for a study article - not a precision rail, not a loose one. 4. **Do you want one probe point on the switching node?** That means a small pad on the switch node with a ground pad right beside it, so a scope probe with a short ground can measure switch-node ringing and efficiency properly at bring-up. This is the one place the mode's boundary needs an owner ruling: the mode excludes test points "beyond the block's own measurement need", and for a board whose purpose is to be studied - and whose bring-up evidence is what proves out the knowledge records it exists to seed - that need is arguable either way. The tradeoff is real: a pad hangs extra copper on the noisiest node on the board. DEFAULT: yes - exactly one switch-node probe pad with an adjacent ground pad, and nothing else. VIN, VOUT and GND are already reachable at the two screw terminals, so no other test points are added.

4 Architecture

Decisions of record (state.json)

Phase	What	Why
P0	Build mode: ultra-bare-bones (token 'ultra bare bones design:' opens the brief)	reference/build-modes.md scope contract: ONE functional block (24V->5V/2A buck) plus only datasheet-required support; one input + one output interface (screw terminals); no protection/filtering/indicators/test points beyond the block's own need; defaults = fewest honest layers, JLC PCBA single-sided, qty 5, bench 0-50C, stop at P9. Research mandatory: first coverage call of each phase runs –maturity-floor proven –research-provisional.
P0	Input ceiling: 30 V is a HARD maximum operating input; converter chosen with Vin absolute-max >= 36 V class (˜20% headroom) as selection headroom only - no added clamp parts	Owner answered P0 Q1 with the default. Keeps VIN-GND at/below 30 V so check_creepage stays a no-op and IPC-2221 spacing does not bind the layout; the mode excludes an input clamp so the IC rating is the only defense.
P0	No live hot-plug of the input: wires land first, supply ramps from zero	Owner answered P0 Q2 with the default. Removes the ˜2x lead-inductance ringing case (˜60 V from 30 V) from the requirement set; datasheet-required bulk input capacitance damps the incidental case, no damping component added.
P0	Output spec: +/-3% DC (4.85-5.15 V) over 18-30 Vin and 0-2 A load; ripple <= 50 mV peak-to-peak	Owner answered P0 Q3 with the default. Sets the regulation/ripple targets that bind feedback divider tolerance, output capacitance and the P8 sim bounds.
P0	One switch-node probe pad plus an adjacent GND pad; no other test points	Owner answered P0 Q4 with the default. Mode-boundary ruling: the switch node is the block's own measurement need on a study board whose bring-up evidence proves the seeded records. Kept minimal and inside the existing SW polygon to limit added copper on the noisiest node.
P1	P1 research roster: research-power-architect + research-component-scout(buck-regulator) only. NO research-interface-spec, NO research-reference-design.	Screw terminals are bound by no electrical standard, so there is no interface spec to research. The buck block is not novel - the library already holds 16 buck records - and the useful vendor-layout digest is exactly what the P3 part-level coverage research produces for the CHOSEN IC; a P1 reference-design spawn would duplicate it. Mode says do not pad.
P1	Topology: synchronous integrated-FET buck, Fsw ˜500 kHz (band 350-600), L = 10 uH	research-power-architect numbers: async costs +0.42 W (+33%) and adds a 0.77 W diode hot spot at Tj ˜126 C against a 125 C rating, worst-cased by D=0.17 (diode conducts 83% of every cycle at 30 V). Fsw >= 1 MHz costs 1.58-1.93 W and collides t_on(30 V) with typical min-on-time. LDO is 38 W - arithmetic. Controller+FETs is the ladder's escape hatch for corners with no stocked integrated part; 30 V/2 A is not one.

Phase	What	Why
P1	PROVISIONAL 2-layer stackup, conditional on the P3 part: sync + Rds_LS <= 85 mOhm + outline >= ~1000 mm2 + >= 16 thermal vias + Tj 150 C	The 2L screen passes by only 2 C (0.92 W x 73.8 C/W = 67.9 vs check_thermal dt_c 70). The layer count FOLLOWS THE PART - a 2 A-class part at Rds_LS 130 mOhm gives 1.19 W and forces 4L. P3 must re-run this arithmetic with the chosen part's real Rds before P5 fixes the stackup (no outline-shrink step exists; board_init is final).
P1	OPEN CONSTRAINT carried to P3 (not settled at P1): light-load ripple vs A3. A3 binds <= 50 mVpp across the FULL 0-2 A range INCLUDING no load, but PFM/burst light-load ripple is typically 50-100 mVpp at 5 V.	research-power-architect OPEN item. The mode excludes the usual fix (a preload resistor is excluded conditioning), so P3 part selection must supply ONE of: a forced-CCM part, a strappable MODE/FCCM pin, or a datasheet figure showing <= 50 mVpp at light load. If no JLC-stocked 36 V-class candidate offers any of the three, ESCALATE to the owner to choose between relaxing A3 at light load and admitting a preload against the mode. A burst-mode part must NOT ship silently against A3.
P2	SKIPPED the optional coverage-mapper spawn on the P2 mapping_request	Every one of the 15 buck records is ALREADY matched to block:B1 via keys and evaluated per class; the blockers are maturity-below-floor (8 classes), envelope-outside (inrush on source_kind, snubber on rectifier_kind, COT on control_kind) and level-below-min (1). A mapping edge cannot change any of those three verdicts, so a mapper pass is a provable no-op here. The only unkeyed library record (spiral-inductor-plane-and-heatsink-keepout) is also 'approved' and would still be maturity-blocked. Recipe step marks the mapper optional.
P2	D5 OUTLINE 40 x 30 mm (1200 mm2) - FINAL, binds at P5 board_init	The outline IS the radiator: R_ba is 39/34/31 C/W at 875/1064/1200 mm2, so area buys more Tj than layers (~10 C vs ~11 C) and is free at the fab. 20% above P1's ~1000 mm2 floor. The increment over 38x28 is bought by physical parts - a 10-12.5 mm inductor (D11), two ~10 mm screw terminals, 4x M3 with 6.5 mm keepouts - not padding. No outline-shrink step exists later. J1 LEFT edge, J2 BOTTOM edge.
P2	D4 STACKUP JLC2313_1.6, 2 layers, 1 oz - CONDITIONAL	Conditions: sync + P_U1 <= 0.95 W at 30V/2A + outline >= 1000 mm2 with unbroken B.Cu pour + >= 16 thermal vias + Tj_max 150 C. Named escalation trigger: P_U1 > 0.95 W (at 400 kHz = Rds_LS > ~110 mOhm at 25 C; at 500 kHz, 85 mOhm) OR a 125 C part OR an async part OR check_thermal erroring at dt_c 70 -> 4L JLC04161H-1080B, outline unchanged. P3 MUST re-run this with the chosen part's real datasheet Rds BEFORE P5 freezes the stackup.

Phase	What	Why
P2	D2 Lead part class: 36 V-class SYNC integrated-FET 3 A-class, exposed-pad SOIC-8-EP, fixed 400 kHz, internally compensated, peak-current-mode (LMR33630ADDAR). ADJUSTABLE output accepted.	CONFLICT RESOLVED: P1 preferred a FIXED 5 V part to delete the divider tolerance term; the scout's JLC sweep found no fixed 5 V sync 36 V-class part, and its only fixed candidate (XL1509-5.0E1) is +/-4 % guaranteed, which alone breaks A3. P1's PREFERENCE loses to stock reality - it was a preference, not a filter. Scout ranks 2-4 are asynchronous and are NOT drop-in fallbacks (D1 rejects async with numbers). If P3 finds a stocked fixed 5 V sync >= 36 V part, take it: R1/R2, the tolerance budget and the only sim candidate all disappear.
P2	D3 fsw 400 kHz, L = 15 uH (P1 recommended 500 kHz / 10 uH)	The lead part is a FIXED 400 kHz device with no RT pin, and 400 kHz sits inside P1's own 350-600 band. At 400 kHz L must be 15 uH (10 uH gives 52% ripple); P_U1 improves 0.924 -> 0.854 W. dI 0.694 A, I_L,pk 2.35 A, ripple 10.4 mVpp. Reverses if the part's internal-compensation L/C table excludes 15 uH - fallback 10 uH at 400 kHz (15.6 mVpp, I_L,pk 2.52 A, both in spec).
P2	D8 canonical net names +VIN /SW +5V GND (/FB and /BST deliberately NOT in constraints power[])	P1 proposed VIN/SW; renamed to match the KiCad export mechanism (power symbols export bare, root-sheet local labels export with one leading slash). A power[] entry on /FB would put a width rule on a high-impedance sense node and pull it into the check_pdn decoupling inventory. P4 must run netlist_audit.py - a rename surfaces as missing_net (error) and nothing else catches it.
P2	D9 ONE FLAT ROOT SHEET, hierarchy rejected; D6 FB divider 0.1 %/25 ppm; D12 exactly ONE sim candidate (FB divider DC accuracy vs A3's window)	D9: 14 parts / 6 nets, and a child-sheet label exports as /<sheet>/<LABEL>, silently unhooking every constraints entry spelling it /<LABEL> (the class that cost lumina-par). D6: 1 % resistors give a worst-case SUM of ~3.7 % which fails A3; 0.5 % closes only on RSS; 0.1 %/25 ppm closes on the worst-case SUM (~2.5 %) for cents and no part-count change. D12: buck switching is not simulated by policy; the FB divider sweep catches the wrong-value class ERC/DRC/DFM are blind to, and disappears if P3 lands a fixed-output part.

Sheet plan

sheets.md - bb-buck schematic sheet plan, refdes ranges, CANONICAL net names

1. Sheet plan: ONE FLAT ROOT SHEET

```
| Sheet | File | Blocks it carries | Interface nets at its boundary | Refdes
ranges | ‘pwr_base‘ |
|---|---|---|---|---|---|
| root (only) | ‘kicad/bb-buck.kicad_sch‘ | B1 buck converter, J1 input, J2
output, TP1/TP2 probe, H1-H4 mounting | none - the board’s only interfaces are
the two physical terminals (‘+VIN‘/‘GND‘ at J1, ‘+5V‘/‘GND‘ at J2) | ‘U1-U9‘,
‘L1-L9‘, ‘C1-C19‘, ‘R1-R9‘, ‘J1-J9‘, ‘TP1-TP9‘, ‘H1-H9‘ | **1** (‘#PWR01..‘,
```

‘#FLG01..‘) |

A hierarchical split is deliberately REJECTED on this board. Hierarchy exists to keep large schematics readable and to give each sheet its own refdes/#PWR range; this board has 14 electrical parts and 6 nets on one A4 page. The cost of splitting is concrete and has been paid before: a label inside a child sheet exports as `/<sheet>/<LABEL>`, which silently unhooks every `constraints.json` entry that spells it `/<LABEL>` (the P4 amendment class that cost `lumina-par`, and the reason `sbuck-5v3a` also went flat). With one root sheet there are no sheet prefixes anywhere, by construction.

Refdes ranges are therefore trivially unique. They are stated anyway because the rule is "unique ACROSS sheets": **if a later amendment ever splits this sheet, the child sheets take `pwr_base = 40` and `80` and keep the prefix ranges above**, and every net name in s2 must be re-checked against the new export.

2. CANONICAL NET NAMES - these are now binding

`research/power.json` proposed `VIN / SW / +5V / GND`. **P2 makes them canonical here, and two of the four change spelling.** These are the names the schematic MUST produce and the names `constraints.json` uses.

Canonical net	KiCad mechanism	Spans	Declared in constraints.json as
‘+VIN‘	power symbol (VALUE = net name) -> exports BARE	J1.1 -> C1/C2/C3 -> U1 VIN pin	‘power‘ 1.1 A, ‘voltages‘ 30 V
‘GND‘	power symbol -> exports BARE	everything; B.Cu pour	‘power‘ 2.6 A ‘plane_fed‘, ‘planes‘ F.Cu + B.Cu, ‘thermal‘ heatsink net
‘+5V‘	power symbol -> exports BARE	L1 -> C4/C5 -> R1 -> J2.1	‘power‘ 2.6 A
‘/SW‘	root-sheet LOCAL LABEL -> exports with ONE leading slash	U1 SW pin -> L1 -> TP1	‘power‘ 2.6 A ‘pdn:false‘, ‘high_speed‘, ‘voltages‘ 30 V
‘/FB‘	root-sheet local label	R1/R2 midpoint -> U1 FB pin	**deliberately NOT declared** (see s4)
‘/BST‘	root-sheet local label	U1 BOOT pin -> C6 -> ‘/SW‘	**deliberately NOT declared** (see s4)

Why `+VIN` and not `VIN`: it is a rail feeding a power pin and decoupled by the input bank, so it is a power symbol, and a power symbol's exported name is bare. Why `/SW` and not `SW`: it is a root-sheet local label, and KiCad prefixes those with a slash on export. **A power symbol WINS over a coincident label**, so P4 must not mix the two mechanisms on one node.

Enforcement, not hope: at P4 run `netlist_audit.py --net kicad/bb-buck.net --constraints architecture/constraints.json`. A net spelled differently in the schematic surfaces as `missing_net` (error). Without that run, a misspelling is silent: `check_current` finds no copper on a net that does not exist and reports nothing.

3. Contents of the root sheet

Refdes	Part	Net attachments
‘U1‘	36 V-class synchronous integrated-FET buck, 3 A-class, exposed pad, 400 kHz, internally compensated	VIN=‘+VIN‘, GND/PGND/EP=‘GND‘, SW=‘/SW‘, BOOT=‘/BST‘, FB=‘/FB‘, EN per datasheet (see below)
‘C1‘	100 nF 50 V X7R - HF input bypass, **‘role: reg_input‘**	‘+VIN‘ / ‘GND‘
‘C2‘ ‘C3‘	10 uF 1210 50 V X7R - input bank	‘+VIN‘ / ‘GND‘
‘C6‘	100 nF - bootstrap (value/rating per datasheet)	‘/BST‘ / ‘/SW‘
‘L1‘	15 uH shielded/composite, <= 40 mOhm, Isat >= ~6.6 A	‘/SW‘ / ‘+5V‘
‘C4‘ ‘C5‘	22 uF 1206 25 V X7R - output bank	‘+5V‘ / ‘GND‘

‘R1‘ ‘R2‘	FB divider, 0.1 % / 25 ppm (‘R1‘ = top, ‘+5V‘->‘/FB‘; ‘R2‘ = bottom, ‘/FB‘->‘GND‘)	as listed
‘J1‘	2-pole 5.08 mm screw terminal, THT	1=‘+VIN‘, 2=‘GND‘
‘J2‘	same part as J1	1=‘+5V‘, 2=‘GND‘
‘TP1‘	SW probe pad (A4) - INSIDE the SW copper, not on a spur	‘/SW‘
‘TP2‘	GND probe pad, adjacent to TP1	‘GND‘
‘H1‘-‘H4‘	M3 clearance holes, 3.2 mm	mechanical (no net)

EN: read the part's own behaviour before adding anything. Most parts in this class auto-start from an internal pull-up, so EN ties to `+VIN` or floats and NO parts are added. No UVLO divider - mode-excluded, and a bench supply has no cable-droop motorboating case (0.62 A over 2 m of lead is ~0.1 V). No soft-start components: internal soft-start makes Cout inrush `C x Vout/tss` ~ 65 mA, and A2 means the supply ramps from zero anyway.

If the FB pin has no divider because P3 found a FIXED 5 V part, `R1`/`R2` and `/FB` disappear and the sense point ties straight to the output - **that is the preferred outcome and it removes the whole s5 tolerance budget**, but do not synthesise it from an adjustable part's datasheet figure.

4. Two nets deliberately left OUT of `constraints.json`

`/FB` and `/BST` are declared nowhere. This is intentional:

- `rules_gen` buckets every `power` entry into a netclass **at that net's own IPC-2152 width** (this used to flatten every power net to the widest width; the current script buckets per net - verified in `rules_gen.net_classes`). Even so, a `power` entry on `/FB` would give the feedback node a minimum-width rule and pull it into the `check_pdn` decoupling inventory, and `/FB` is a high-impedance sense node that wants to be SHORT and THIN and nowhere near `/SW`.
- `/BST` carries only gate charge. Nothing to size, nothing to decouple.

The two rules that DO apply to them are layout rules, carried in s5 below and in `constraints.json placement.groups` / `separation`, not width rules.

5. Placement groups this sheet implies (they become P6 groups)

- **‘hotloop‘** (anchor `U1`; `C1`, `C2`, `C3`, `C6`): C1 nearest the VIN pin, then the bulk ceramics, then U1 VIN -> U1 PGND, on F.Cu, **smallest enclosed area achievable**. This is the loop that spends ~37 mV of the 50 mV ripple budget. It must be placed by explicit `place_edit` and LOCKED before `place_anneal` runs - an annealer optimises a cost function and cannot discover a hot loop.
- **‘output‘** (anchor `L1`; `C4`, `C5`): Cout ground and Cin ground return to the GND pour at SEPARATE points near their own pins - never a shared narrow neck (`buck-cin-co-ground-separation`).
- **‘feedback‘** (anchor `R1`; `R2`): at the FB pin, sense point AFTER the output caps, >= 5 mm from L1 and off the `/SW` side of the package.
- **‘probe‘** (anchor `TP1`; `TP2`): TP1 inside the existing `/SW` pour; TP2's ground returns straight down to the B.Cu pour beneath it.

`/SW` itself: a short WIDE pour, 1.6-2.0 mm wide, <= 8 mm long, <= 40 mm^2 total, F.Cu only, never near a board edge or a screw terminal. IPC-2152 constrains the WIDTH (a floor); EMI constrains the AREA (a ceiling that nothing in the toolchain enforces). TP1's pad counts against that 40 mm^2.

Full architecture narrative (not inlined; see the artifact index): `architecture/blocks.md`, `architecture/decisions.md`, `architecture/power_tree.md`, `architecture/stackup.md`.

5 Schematic

Pending — produced at P4.

6 Layout

Pending — produced at P6.

7 Verification

Pending — produced at P8.

8 DFM and Fabrication

Pending — produced at P9.

9 Run Record (Appendix)

Phase digests

P0

```
# P0 Intake - digest (2026-08-15)

- bb-buck:  non-isolated buck, 18-30 V DC in (24 nom) -> 5 V @ 0-2 A out, screw
terminal both ends, bench supply + resistive load.
- Mode **ultra-bare-bones** recorded as a decision:  buck block +
datasheet-required
support only.  Defaults applied (qty 5, JLC PCBA single-sided, 0-50 C bench, no
enclosure, fewest honest layers, smallest honest outline, stop at P9).
- requirements.md by ‘requirements-analyst‘; ‘check_requirements.py‘ PASS 9/9,
0
violations, before and after fold-in.
- Section 8 closed with NO safety question open, derived:  bench source (no
battery),
resistive load (no motor), ~2.5 A peak (< 3 A), 30 V is not >30 V and < 60 V
ES1.
- 4 questions asked in ONE batch, owner took all four defaults -> now
requirements:
A1 30 V hard max operating, part abs-max >= 36 V (headroom in the PART, no
clamp);
A2 no live hot-plug (drops the ~60 V ring case); A3 +/-3 % DC over full
line+load,
<= 50 mV pk-pk; A4 one switch-node probe pad + adjacent GND pad, nothing else.
- Left open for P1/P2:  sync vs async rectification; 2L vs 4L (2L assumed,
revisable
on thermal grounds at ~0.8-1.8 W / 50 C ambient).
- Fable tier out of credits -> fable/high spawns run opus/xhigh (ledgered).
```

P1

```
# P1 Research - digest (2026-08-15)

- Roster:  power-architect + component-scout(buck-regulator) only.  No
interface-spec
(screw terminals bind no standard), no reference-design (P3 part-level coverage
research produces the vendor layout digest for the CHOSEN IC instead).
```

- Topology ruled **synchronous integrated-FET, ~500 kHz (350-600), L = 10 uH**. Async
rejected: +0.42 W (+33 %) plus a 0.77 W diode at Tj ~126 C vs a 125 C rating (D=0.17
means it conducts 83 % of every cycle at 30 V). >= 1 MHz rejected: 1.58-1.93 W and
t_on(30 V) collides with min-on-time.
- Part envelope for P3: Vin abs-max >= 36 V, **Rds_LS <= 85 mOhm (rank on LS not HS+LS;
at D=0.167 the LS FET conducts 5x longer)**, t_on_min <= 130 ns, Tj 150 C, exposed pad
mandatory, P_U1 <= 0.95 W at 30 V/2 A. A 3 A-class part, not a marginal 2 A one.
- **2L CONDITIONAL** - the screen passes by 2 C (0.92 W x 73.8 C/W = 67.9 vs dt_c 70).
Needs sync + Rds_LS <= 85 mOhm + outline >= ~1000 mm^2 + >= 16 thermal vias + Tj 150 C.
Layer count FOLLOWS THE PART; re-run at P3 before P5 fixes the stackup. Mode note:
"smallest honest outline" has a numeric floor - the outline IS the radiator.
- Scout, 5 candidates, every figure datasheet-sourced: top **LMR33630ADDAR** C841384
(sync, 38 V, 400 kHz, $0.74@5, stock 6885); AOZ1284PI C48060 async 40 V $0.85;
TPS54560BDDAR C1850354 65 V $1.06. AP63357Q out at 35 V abs-max (1 V short of A1);
LMR33630's 1.4/2.1 MHz siblings pulse-skip at 30 V - only the 400 kHz "A" clears it;
XL1509 is the only Basic part but its own +/-4 % already breaks A3's +/-3 %.
- Convergence: the scout's top pick sits inside the architect's envelope.

P2

P2 Architecture - digest (2026-08-15)

- One block (B1 sync buck) + J1/J2 screw terminals + TP1/TP2 SW probe pair + 4x M3.
14 parts, 6 nets, ONE FLAT ROOT SHEET (a child-sheet label exports as
/<sheet>/<LABEL> and silently unhooks constraints - the class that cost lumina-par).
Canonical nets now binding: +VIN /SW +5V GND (/FB, /BST undeclared by design).
- Stackup JLC2313_1.6, 2L, 1 oz - CONDITIONAL, escalation trigger named numerically
(P_U1 > 0.95 W = Rds_LS > ~110 mOhm at 400 kHz). Outline **40 x 30 mm FINAL**.
- fsw 400 kHz / L 15 uH (part is fixed-400 kHz, inside P1's band). P1's fixed-output
preference LOST to stock reality -> adjustable + FB divider at 0.1 %/25 ppm, which
closes A3 on worst-case SUM (2.5 %) not just RSS. constraints_lint 0/0 (verified).
- Architect corrected 3 premises in its own brief, all verified: rules_gen already
buckets power nets per IPC-2152 width (no flattening defect); planes[] REPLACES
layer defaults and rejects unknown keys incl. `_note`; `pdn:false` also skips
check_irdrop/check_pdn_z so it was dropped from GND. -> LEARNINGS.md entry.
- **Coverage (mode floor `proven` + --research-provisional): 1 slot, 10 classes, all
gap** - 8 blocked only on maturity, 3 substantive (inrush outside on source_kind,
snubber outside on rectifier_kind, COT outside on control_kind). Mapper skipped as

```
a provable no-op (every record already keyed; edges cannot move
maturity/envelope).
- Research task block-buck-1:  3 sources (LMR33630 SNVSAN3F vendor-layout + 2
TI app
notes), 12+ pages read visually, **9 records, 9 verified / 0 refuted**, depth
3/4.
Second reader REFUTED one on first pass (record read Li as 0.3 uH; the figure
prints
**9.3 uH** - a 31x error that mis-sized the damping criterion).  Corrected,
re-read
fresh-context with gridline-calibrated vector measurement, verified.
- **Re-run coverage:  9 covered / 1 provisional (inrush) / 0 gap, exit 0.**
Research
narrowed the only genuine library hole; the owner's promotion ruling closes it.
- Design-relevant findings:  (a) hot-plug at 30 V would ring to 45-49 V vs the
part's
38 V abs-max - A2 (no live hot-plug) is what makes the clamp-free mode safe;
(b) the EP is **AGND**, the reference all electrical parameters are measured
to, so
the thermal via array is a loop-accuracy item; (c) 2L is a DOCUMENTED deviation
from
the datasheet's own layout rule 5, measured ~5 dBuV/m - binds P6/P7 to keep the
bottom pour unbroken under hot loop, SW and inductor.
```

Gate attempt history

(no entries)

Issues

(no entries)

Human checkpoints

Id	Phase	Status	When	Note
1	P2	not reached		
2	P4	not reached		
3	P6	not reached		
4	P8	not reached		
5	P10	not reached		

10 Artifact Index

File	Size	Note
`state.json`	16,933 B	
`brief/brief.md`	605 B	
`requirements.md`	16,422 B	
`architecture/blocks.md`	10,091 B	
`architecture/constraints.json`	17,690 B	
`architecture/decisions.md`	9,227 B	
`architecture/power_tree.md`	10,485 B	
`architecture/sheets.md`	7,162 B	
`architecture/stackup.md`	7,324 B	